

Yip Kai Men

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EDUCATION

National University of Singapore <i>Bachelor of Engineering, Mechanical</i> Relevant Coursework: Machine Learning, Data Analysis and Visualization, Probability and Statistics	Aug 2021 - May 2025
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PROFESSIONAL EXPERIENCE

TOFFs Technologies <i>AI Engineer</i> - Architected multi-agent systems using various LLM frameworks and MCP servers, designing state schemas, failure-recovery policies, and human-in-the-loop approval gates alongside RAG and Neo4j knowledge graphs for grounded, auditable agent reasoning. - Built and deployed full-stack AI applications spanning LLM orchestration, REST APIs, and frontend interfaces, owning the complete technical architecture from design through production deployment. - Led end-to-end delivery of AI-assisted engineering tools as primary technical owner, translating domain expert knowledge into structured system specifications and production-ready prototypes. - Managed stakeholder discovery across client leadership, domain experts, and internal supervisors, synthesising requirements into PRD-driven development workflows that scoped MVPs for agentic builds.	May 2025 - Present
Bosch <i>AI Engineer Intern</i> - Utilized MATLAB Simulink for sophisticated modeling and simulation of feedback control systems, providing essential reference data for experimental analysis. - Contributed to research on sensor anomaly cases, assisting in the development of innovative solutions for problem identification and correction. - Conducted comprehensive laboratory experiments to collect test data, integral in training a robust anomaly detection algorithm using machine learning techniques such as K-means clustering and Mahalanobis distance. - Employed various machine learning techniques, including Convolutional Neural Networks (CNNs) on diverse datasets, to refine predictive models and enhance outlier detection.	Jan 2024 - May 2024
Edwards Lifesciences <i>Digital Transformation and Analytics Intern</i> - Conducted time study and data analysis of tissue processing and valve stem assembly timelines and operations, delivering visualized insights to the team. - Implemented impact studies on Manufacturing Execution System (MES) to understand its impact on diverse assembly/processing phases, concluding a 46% increase in administration productivity for 1 manufacturing process. - Effectively set up digital work stations using CAMSTAR MES, expanding production line and improving data flow. - Proficiently participated in User Acceptance Testing and Functional Testing, ensuring seamless software alignment with business requirements for 1 major manufacturing process.	Jul 2023 - Dec 2023

SKILLS

- Languages: English, Mandarin, Cantonese - Software/Programming: Python, R, SQL, MATLAB, Tableau, Power BI, Knowledge Graphs, GNNs, Microsoft Office, SOLIDWORKS, CSS, HTML	
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PROJECTS & COMPETITIONS

HSBC Global Private Banking Case Competition 2023 <i>Participated in a team of 4 to construct and present our specialized pitch on asset management and market insights</i>	Oct 2023
SPY Analytics and Forecasting <i>A deep dive into SPY's historical performance, inflation-adjusted returns, and future growth potential using machine learning</i>	Aug 2024 - Dec 2024
SMU Business Intelligence and Analytics Club Datathon 2025 <i>Delivered interactive dashboards containing with extracted structured data on Plotly Dash for visualisation</i>	Jan 2025 - Feb 2025

CERTIFICATES

Machine Learning Specialization, DeepLearning.AI	Jun 2023
Applied Machine Learning in Python, University of Michigan	Jul 2023
Using SQL Databases with Python, University of Michigan	Jul 2023
AWS Cloud Practitioner Essentials, Amazon Web Services (AWS)	Aug 2023
Advanced Data Analytics Specialization, Google	Sep 2024